s2_8_NET_L22_ChatPDF_S_CoT_AU2440276_Manthai

L21_L22.pdf
PDF

You are generating a lecture scribe intended as exam reference material for a closed-notes / reading-based exam.

Authoritative Context (Strict)
The only information you may use is the material explicitly provided as source content for the lecture.
This may include lecture slides, lecture PDFs, textbook excerpts, handwritten notes, or any combination thereof.

The lecture slides or primary lecture document define the structure, ordering, scope, and core content of the lecture and must be fully and faithfully represented in the lecture scribe.

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You must not use outside knowledge.
You must not invent definitions, notation, results, proofs, derivations, examples, or assumptions.

Prompting Style: Structured Reasoning (with Chain-of-Thought)

Use careful, step-by-step reasoning internally to construct the lecture scribe.

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Produce a faithful, structured reconstruction of what was taught in the lecture, such that a student who studies only this document can reliably revise the material for the exam.

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Prohibitions

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Final Output

A clean, logically organized lecture scribe suitable for direct exam preparation.

Below is the **Lecture 21–22 Scribe: Inequalities and Tail Bounds**, reconstructed strictly from the provided lecture slides and material L21_L22.

Lecture 21–22: Inequalities and Tail Bounds

1. Tails

Definition

The **tail of a random variable** X is:

$$P(X > x)$$

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)

Why are tails hard to compute?

- Requires summing many small probabilities

---

2. Tail Examples

Example 1: Load Distribution Across Servers

Problem
Suppose distributing n jobs among n servers at random.
Find:

$$P(\text{a particular server gets } \geq k \text{ jobs})$$

Model

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Observation

- Most servers get few jobs
- Occasionally one server gets many jobs
- This is a **tail event (rare overload)**

Example 2: Poisson Arrivals

Problem
Jobs arrive with rate λ jobs/hour.
Find:

$$P(X \geq k)$$

Model

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Observation

- Most hours have moderate arrivals
- Occasionally, bursts occur
- This is a **tail event (sudden spike in load)**

---

3. Why Tail Bounds?

Exact computation

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

- These are **complicated long sums**

Idea

Instead of exact value:

- Find an **upper bound**

- Easier to compute
- Still gives a **guarantee**

$$P(X \geq k) \leq \text{simple bound}$$

Key Idea

Approximate safely instead of computing exactly

---

4. Running Example

- Flip a fair coin n times
- Let $X =$ number of heads

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

$$\mu = E[X] = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3n}{4}\right)$$

---

5. Markov's Inequality

Key Idea

- Uses only the mean
- Large values are rare

Statement

For non-negative X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (Idea)

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

Consider only $x \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

Replace x by a:

$$\geq \sum_{x=a}^{\infty} a \cdot p_X(x)$$

$$= a \cdot P(X \geq a)$$

Thus:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Running Example

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad E[X] = \frac{n}{2}$$

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{n/2}{3n/4} = \frac{2}{3}$$

Observation:

- Bound does not depend on n
- Weak (loose) bound

Real-Life Example: Income Distribution

- Average income = 10,000

$$P(X \geq 50,000) \leq \frac{10,000}{50,000} = 0.2$$

Conclusion

- At most 20% earn ≥ 50,000

Solved Example 1: Exam Grades

- $\mu = 70$, threshold $a = 90$

$$P(X \geq 90) \leq \frac{70}{90} = \frac{7}{9}$$

$$P(X \geq 90) \leq 0.778$$

Solved Example 2: Server Load

- $E[X] = 5000$, threshold $a = 20000$

$$P(X \geq 20000) \leq \frac{5000}{20000} = \frac{1}{4}$$

$$P(X \geq 20000) \leq 0.25$$

---

6. Chebyshev's Inequality

Key Idea

- Uses mean + variance
- Controls deviation from mean

Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof Idea

Apply Markov to $(X - \mu)^2$:

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$

$$\leq \frac{E[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

Running Example

$$X \sim \mathrm{Binomial}(n, \tfrac{1}{2})$$

$$\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$$

$$P\left(X \geq \frac{3n}{4}\right) = P\left(X - \frac{n}{2} \geq \frac{n}{4}\right)$$

$$\leq P\left(|X - \mu| \geq \frac{n}{4}\right) \leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Key Insight

- Bound decreases as n increases
- Better than Markov

Real-Life Example: Exam Scores

- Mean = 70, std dev = 10

$$P(|X-70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

Solved Example 1

- $\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25$
- Threshold: $X \geq 90 \Rightarrow |X-70| \geq 20$

$$P(X \geq 90) \leq P(|X-70| \geq 20) \leq \frac{25}{400} = \frac{1}{16}$$

$$P(X \geq 90) \leq 0.0625$$

Solved Example 2: Network Latency

- $\mu = 50, \sigma^2 = 16$
- Outside [30, 70] ⇒ $|X-50| \geq 20$

$$P(|X-50| \geq 20) \leq \frac{16}{400} = \frac{1}{25}$$

$$P(|X-50| \geq 20) \leq 0.04$$

---

7. Chernoff Bound

Progression

- Markov → uses X
- Chebyshev → uses $(X-\mu)^2$
- Chernoff → uses e^{tX}

Key Insight

- Exponential grows very fast
- Large values amplified
- Gives tighter tail bounds

Core Trick

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Final Form

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight

- Holds for all $t > 0$
- Choose best $t \to$ tightest bound

Strength

- Much stronger than Markov & Chebyshev
- Gives **exponential decay**

---

8. Chernoff Bound (Lower Tail)

Goal

$$P(X \leq a)$$

Trick

Use $t < 0$:

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

Final Form

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

---

9. Chernoff for Poisson

Goal

$$X \sim \text{Poisson}(\lambda)$$

Compute:

$$E[e^{tX}]$$

Derivation

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \cdot e^{-\lambda} \frac{\lambda^i}{i!}$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

End of Lecture 21–22 Scribe

If you want, I can now:

- Convert this into **Overleaf (LaTeX) code**, or
- Make a **super-condensed revision sheet (1–2 pages)** for exam use.

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Final Output

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Below is the **faithful, structured lecture scribe for Lecture 21–22**, reconstructed strictly from the provided lecture material, preserving order, notation, derivations, and worked steps exactly as presented L21_L22.

Lecture 21–22: Inequalities and Tail Bounds

---

1. Tail of a Random Variable

Definition

The tail of a random variable X is:

$$P(X > x)$$

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)

Why are tails hard to compute?

- Requires summing many small probabilities

---

2. Tail Examples

Example 1: Jobs Across Servers

Problem
Suppose n jobs are distributed among n servers at random.
Find:

$$P(\text{a particular server gets } \geq k \text{ jobs})$$

Model

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Observation

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is a **tail event (rare overload)**

Example 2: Poisson Arrivals

Problem
Jobs arrive according to a Poisson process with rate λ jobs/hour.
Find:

$$P(X \geq k)$$

Model

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Observation

- Jobs arrive randomly over time

- Most hours have moderate arrivals
- Sometimes there is a burst of arrivals

This is a **tail event (sudden spike in load)**

---

3. Why Tail Bounds?

Exact Computation

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

- Complicated, long sums

Idea

Instead of exact value:

- Find an **upper bound**
- Easier to compute
- Still gives a **guarantee**

$$P(X \geq k) \leq \text{something simple}$$

Key Idea

Approximate safely instead of computing exactly

---

4. Running Example (Key Idea)

- Flip a fair coin n times

Distribution

$$X = \text{number of heads}, \quad X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean

$$E[X] = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3n}{4}\right)$$

---

5. Markov's Inequality

Key Idea

- Uses only the mean
- Large values are rare

Statement

For non-negative X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (as presented)

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

Consider only $X \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

Replace x by a:

$$\geq \sum_{x=a}^{\infty} a \cdot p_X(x)$$

$$= a \cdot P(X \geq a)$$

Thus:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Running Example Application

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad E[X] = \frac{n}{2}$$

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{n/2}{3n/4} = \frac{2}{3}$$

Observation

- Does not involve n
- Weak bound

Real-Life Example: Income Distribution

- Average income = 10,000

$$P(X \geq 50,000) \leq \frac{10,000}{50,000} = 0.2$$

At most 20% people earn 50,000

Solved Example 1: Exam Grades

Given

- $\mu = 70$, threshold $a = 90$

Steps

Step 1:

$$X \geq 0, \quad E[X] = 70$$

Step 2:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 3:

$$P(X \geq 90) \leq \frac{70}{90}$$

Step 4:

$$= \frac{7}{9}$$

Result

$$P(X \geq 90) \leq 0.778$$

Solved Example 2: Server Load

Given

- $E[X] = 5000$, threshold $a = 20000$

Steps

Step 1:

$$X = \text{requests/sec}, \quad E[X] = 5000$$

Step 2:

$$a = 20000$$

Step 3:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 4:

$$P(X \geq 20000) \leq \frac{5000}{20000}$$
$$= \frac{1}{4}$$

Result

$$P(X \geq 20000) \leq 0.25$$

---

6. Chebyshev's Inequality

Key Idea

- Uses mean and variance
- Controls deviation from mean

Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof Idea (as presented)

Apply Markov to $(X - \mu)^2$:

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$
$$\leq \frac{E[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

Running Example

$$X \sim \mathrm{Binomial}(n, \tfrac{1}{2})$$

$$\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$$

$$P\left(X \geq \frac{3n}{4}\right) = P\left(X - \frac{n}{2} \geq \frac{n}{4}\right)$$

$$\leq P\left(|X - \mu| \geq \frac{n}{4}\right)$$

$$\leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Real-Life Example: Exam Scores

- Mean = 70, std deviation = 10

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

Solved Example 1

Given

- $\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25$

Step 1:

$$|X - 70| \geq 20$$

Step 2:

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

Step 3:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Step 4:

$$P(|X - 70| \geq 20) \leq \frac{25}{400}$$

$$= \frac{1}{16}$$

Result

$$P(X \geq 90) \leq 0.0625$$

Solved Example 2: Network Latency

Given

- $\mu = 50, \sigma^2 = 16$

Step 1:

$$X \notin [30, 70] \Rightarrow |X - 50| \geq 20$$

Step 2:

$$P(|X - 50| \geq 20) \leq \frac{16}{400}$$

Step 3:

$$= \frac{1}{25}$$

Result

$$P(|X - 50| \geq 20) \leq 0.04$$

---

7. Chernoff Bound

Progression of Ideas

- Markov: bound using X
- Chebyshev: bound using $(X - \mu)^2$
- Chernoff: bound using e^{tX}

Core Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov on e^{tX}

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

- $e^{tX} \geq 0 \Rightarrow$ Markov applies

Final Form

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight

- Holds for all $t > 0$

- Choose t to minimize bound

Strength

- Much stronger than Markov and Chebyshev
- Exponential decay

---

8. Chernoff Bound (Lower Tail)

Goal

$$P(X \leq a)$$

Transformation

Use $t < 0$:

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

Final Form

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

---

9. Chernoff Bound for Poisson Tail

Goal

$$X \sim \text{Poisson}(\lambda)$$

Compute $E[e^{tX}]$

Derivation

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \cdot e^{-\lambda} \frac{\lambda^i}{i!}$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

End of Lecture 21–22

If you want next, I can convert this into:

- **Exact Overleaf (LaTeX code matching this structure)**
- Or a **last-day revision sheet (ultra compressed, exam-ready)**

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Final Output

A clean, logically organized lecture scribe suitable for direct exam preparation.

Below is the **complete, faithful lecture scribe for Lecture 21–22**, strictly reconstructed from the provided lecture slides, preserving order, notation, derivations, and all intermediate steps as presented.
L21_L22

Lecture 21–22: Inequalities and Tail Bounds

---

1. Tail of a Random Variable

Definition

The tail of a random variable X is:

$$P(X > x)$$

(*Definition shown in slide "Tails"*)
L21_L22

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer
  (*Slide: Tails – motivation*)
L21_L22

Key Idea

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)
  (*Slide explicitly states this comparison*)
L21_L22

Why are tails hard to compute?

- Requires summing many small probabilities
  (*Explicit statement in slide*)
L21_L22

---

2. Tail Examples

Example 1: Jobs Across Servers

Problem
Suppose you're distributing n jobs among n servers at random.
What's the probability that a particular server gets $\geq k$ jobs?
L21_L22

Model

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Observations (from slide)

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

$$\text{This is a tail event (rare overload)}$$

Example 2: Poisson Arrivals

Problem
Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour.
What's the probability that $\geq k$ jobs arrive during the first hour?
L21_L22

Model

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Observations (from slide)

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes, we see a burst of arrivals

$$\text{This is a tail event (sudden spike in load)}$$

---

3. Why Tail Bounds?

Exact Computation

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Observation

- Complicated, long sums

Idea

Instead of exact value:

- Find an **upper bound**
- Easier to compute
- Still gives a **guarantee**

$$P(X \geq k) \leq \text{something simple}$$

Key Idea

Approximate safely instead of computing exactly
(*Explicit statement on slide*)
L21_L22

---

4. Running Example (Key Idea)

Setup

- Flip a fair coin n times

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

$$E[X] = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3n}{4}\right)$$

---

5. Markov's Inequality

Key Idea

- Uses only the mean
- Large values ⇒ rare
  (*Slide explicitly states*)

L21_L22

Statement

For non-negative X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (as presented in slides)

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

Consider only $x \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

Replace x by a:

$$\geq \sum_{x=a}^{\infty} a \cdot p_X(x)$$

$$= a \cdot P(X \geq a)$$

Thus:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Running Example

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad E[X] = \frac{n}{2}$$

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{n/2}{3n/4} = \frac{2}{3}$$

Observation (slide)

- Does not involve n
- Clearly a terrible bound

Real-Life Example: Income Distribution

$$E[X] = 10{,}000$$

$$P(X \geq 50{,}000) \leq \frac{10{,}000}{50{,}000} = 0.2$$

\text{At most 20% people can earn 50,000}

Solved Example 1: Exam Grades

Given

$$E[X] = 70, \quad a = 90$$

Steps

1. $X \geq 0$, identify $\mu = E[X] = 70$
2. Apply Markov:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

3. Substitute:

$$P(X \geq 90) \leq \frac{70}{90}$$

4. Simplify:

$$= \frac{7}{9}$$

Result

$$P(X \geq 90) \leq 0.778$$

Solved Example 2: Server Load

Given

$$E[X] = 5000, \quad a = 20000$$

Steps

1. Identify mean
2. Define threshold
3. Apply Markov:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

4. Substitute:

$$P(X \geq 20000) \leq \frac{5000}{20000}$$
$$= \frac{1}{4}$$

Result

$$P(X \geq 20000) \leq 0.25$$

---

6. Chebyshev's Inequality

Key Idea

- Uses mean + variance
- Controls deviation from mean

L21_L22

Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof (as presented)

Apply Markov to $(X - \mu)^2$:

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$
$$\leq \frac{E[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

Running Example

$$X \sim \mathrm{Binomial}(n, \tfrac{1}{2})$$
$$\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$$
$$P\left(X \geq \frac{3n}{4}\right) \leq P\left(|X - \mu| \geq \frac{n}{4}\right)$$
$$\leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Observation

- Bound decreases as n increases

- Better than Markov

Real-Life Example: Exam Scores

$$\mu = 70, \quad \sigma = 10$$

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

Solved Example 1

$$\mu = 70, \quad \sigma^2 = 25$$

Steps:

1. $X \geq 90 \Rightarrow |X - 70| \geq 20$
2. Apply Chebyshev:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

3. Substitute:

$$P(|X - 70| \geq 20) \leq \frac{25}{400}$$
$$= \frac{1}{16}$$

Result

$$P(X \geq 90) \leq 0.0625$$

Solved Example 2: Network Latency

$$\mu = 50, \quad \sigma^2 = 16$$

Steps:

1. Outside range ⇒ $|X - 50| \geq 20$
2. Apply Chebyshev:

$$P(|X - 50| \geq 20) \leq \frac{16}{400}$$

3. Simplify:

$$= \frac{1}{25}$$

Result

$$P(|X - 50| \geq 20) \leq 0.04$$

---

7. Chernoff Bound

Progression of Ideas

- Markov: bound using X
- Chebyshev: bound using $(X - \mu)^2$
- Chernoff: bound using e^{tX}

Key Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

- $e^{tX} \geq 0 \Rightarrow$ Markov applies

Final Form

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight

- Holds for all $t > 0$
- Choose best $t \Rightarrow$ tightest bound
- Much stronger than Markov/Chebyshev
- Exponential decay

---

8. Chernoff Bound (Lower Tail)

Goal

$$P(X \leq a)$$

Transformation

Use $t < 0$:

$$P(X \le a) = P(e^{tX} \ge e^{ta})$$

Final Form

$$P(X \le a) \le \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

---

9. Chernoff Bound for Poisson Tail

Goal

Bound tail of:

$$X \sim \text{Poisson}(\lambda)$$

Step 1: Compute $E[e^{tX}]$

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \cdot e^{-\lambda} \frac{\lambda^i}{i!}$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

End of Lecture 21–22

This scribe preserves:

- Exact slide ordering
- All formulas and derivations
- All worked examples with full steps
- All stated observations and results

If you want, I can next:

- Convert this into **perfect Overleaf code (exam-ready formatting)**
- Or create a **1-page ultra-condensed revision sheet**

Sources

L21_L22.pdf
PDF

You are generating a lecture scribe intended as exam reference material for a closed-notes / reading-based exam.

Authoritative Context (Strict)
The only information you may use is the material explicitly provided as source content for the lecture.
This may include lecture slides, lecture PDFs, textbook excerpts, handwritten notes, or any combination thereof.

The lecture slides or primary lecture document define the structure, ordering, scope, and core content of the lecture and must be fully and faithfully represented in the lecture scribe.

Any supplementary material (such as handwritten notes or textbook excerpts) is to be treated as authoritative lecture material intended to:
-> clarify core content
-> add intermediate steps omitted in primary material
-> state assumptions or conditions mentioned during lecture
-> annotate or expand directly corresponding lecture material

If supplementary material contains:
-> clarifications,
-> intermediate steps,
-> assumptions,
-> annotations,
-> or details not explicitly written in the primary lecture document but directly corresponding to it,

they must be integrated at the appropriate location aligned with the primary lecture structure.

Supplementary material must not replace, overshadow, or reorder content that appears in the primary lecture document.

You must not use outside knowledge.
You must not invent definitions, notation, results, proofs, derivations, examples, or assumptions.

Prompting Style: Structured Reasoning (with Chain-of-Thought)

Use careful, step-by-step reasoning internally to construct the lecture scribe.

The lecture scribe must be produced in a single, complete, logically consistent output that strictly follows the provided source material.

Objective

Produce a faithful, structured reconstruction of what was taught in the lecture, such that a student who studies only this document can reliably revise the material for the exam.

The scribe must allow the student to:
-> Recall definitions and notation
-> Provide formulas which are useful for exam prep

-> Reconstruct proofs and derivations
-> Follow worked examples step by step
-> Understand logical dependencies between concepts

Content Requirements (Include Where Applicable)

-> All definitions and notation exactly as introduced
-> All assumptions, conditions, and domains of validity
-> Statements of theorems, propositions, lemmas, or results
-> Proofs or proof sketches reconstructed step by step as presented
-> Worked examples shown in full, with all intermediate steps
-> Explicit logical dependencies between concepts

All primary lecture content must be explicitly represented in the scribe.
Supplementary material may add detail but must not introduce independent structure or unrelated content.

Reasoning Requirement

When a proof, derivation, or example appears in the provided material:
-> Reproduce the explicit logical steps required to follow it
-> Do not add intuition, motivation, alternative arguments, shortcuts, or commentary not present in the provided material

Structure and Style Constraints

-> Follow the same order and scope as the primary lecture document
-> Use precise, formal, exam-appropriate language
-> Be concise but complete; do not omit steps shown in the provided material
-> Do not simplify, re-teach, or editorialize
-> Do not summarize at a high level unless explicitly done in the lecture material

Prohibitions

-> No invented content
-> No external examples
-> No intuition or explanations beyond the provided material
-> No references to prompting, instructions, or the existence of an AI

Final Output

A clean, logically organized lecture scribe suitable for direct exam preparation.

Below is the **complete, strictly faithful lecture scribe for Lecture 21–22**, reconstructed in exact structure, order, and detail from the provided lecture slides, including all definitions, derivations, and

worked examples.
L21_L22

Lecture 21–22: Inequalities and Tail Bounds

---

1. Tail of a Random Variable

Definition

The tail of a random variable X is:

$$P(X > x)$$

(*Definition as shown on slide "Tails"*)
L21_L22

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer
  (*Slide: page 5*)
L21_L22

Key Idea

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)
  (*Slide: page 6*)
L21_L22

Why are tails hard to compute?

- Requires summing many small probabilities
  (*Slide: page 7*)
L21_L22

---

2. Tail Examples

Example 1: Jobs Across Servers

Problem
Suppose you're distributing n jobs among n servers at random.
What's the probability that a particular server gets $\geq k$ jobs?
L21_L22

Model

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Observations

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

$$\text{This is a tail event (rare overload)}$$

Example 2: Poisson Arrivals

Problem
Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour.
What's the probability that $\geq k$ jobs arrive during the first hour?
L21_L22

Model

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Observations

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes, we see a burst of arrivals

$$\text{This is a tail event (sudden spike in load)}$$

---

3. Why Tail Bounds?

Exact Computation

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Observation

- Complicated, long sums
  (*Slide: page 14*)
  L21_L22

Idea

Instead of exact value:

- Find an upper bound
- Easier to compute
- Still gives a guarantee

$$P(X \geq k) \leq \text{something simple}$$

Key Idea

Approximate safely instead of computing exactly
(*Slide: page 15*)
L21_L22

---

4. Running Example (Key Idea)

Setup

Flip a fair coin n times.

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

$$E[X] = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3n}{4}\right)$$

(*Slides: pages 16–17*)
L21_L22

---

5. Markov's Inequality

Key Idea

- Uses only the mean

- Large values ⇒ rare
  (*Slide: page 18*)
  L21_L22

Statement

For non-negative X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof

$$\begin{aligned} E[X] &= \sum_{x=0}^{\infty} x \cdot p_X(x) \\ &\geq \sum_{x=a}^{\infty} x \cdot p_X(x) \\ &\geq \sum_{x=a}^{\infty} a \cdot p_X(x) \\ &= a \cdot P(X \geq a) \end{aligned}$$

Thus:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

(*Slide: page 19*)
L21_L22

Running Example

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad E[X] = \frac{n}{2}$$

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{n/2}{3n/4} = \frac{2}{3}$$

Observation

- Does not depend on n
- Clearly a terrible bound
  (*Slide: page 21*)
  L21_L22

Real-Life Example: Income Distribution

$$E[X] = 10{,}000$$

$$P(X \geq 50{,}000) \leq \frac{10{,}000}{50{,}000} = 0.2$$

\text{At most 20% people can earn 50,000}

(*Slide: page 24*)

L21_L22

Solved Example 1: Exam Grades

Given

$$E[X] = 70, \quad a = 90$$

Steps

1. $X \geq 0$, identify $\mu = 70$
2. Apply Markov:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

3. Substitute:

$$P(X \geq 90) \leq \frac{70}{90}$$

4. Simplify:

$$= \frac{7}{9}$$

Result

$$P(X \geq 90) \leq 0.778$$

(*Slides: pages 25–29*)

L21_L22

Solved Example 2: Server Load

Given

$$E[X] = 5000, \quad a = 20000$$

Steps

1. Identify mean
2. Define threshold
3. Apply Markov:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

4. Substitute:

$$P(X \geq 20000) \leq \frac{5000}{20000}$$
$$= \frac{1}{4}$$

Result

$$P(X \geq 20000) \leq 0.25$$

(*Slides: pages 30–34*)
L21_L22

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6. Chebyshev's Inequality

Key Idea

- Uses mean + variance
- Controls deviation from mean
  (*Slide: page 36*)
L21_L22

Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof

Apply Markov to $(X - \mu)^2$:

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$
$$\leq \frac{E[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

(*Slide: page 38*)
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Running Example

$$\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$$
$$P\left(X \geq \frac{3n}{4}\right) \leq P\left(|X - \mu| \geq \frac{n}{4}\right)$$
$$\leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Observation

- Bound decreases as n increases
- Better than Markov
  (*Slide: page 39*)
  L21_L22

Real-Life Example: Exam Scores

$$\mu = 70, \quad \sigma = 10$$

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

(*Slide: page 42*)
L21_L22

Solved Example 1

$$\mu = 70, \quad \sigma^2 = 25$$

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

$$\leq \frac{25}{400} = \frac{1}{16}$$

Result

$$P(X \geq 90) \leq 0.0625$$

(*Slides: pages 43–47*)
L21_L22

Solved Example 2: Network Latency

$$\mu = 50, \quad \sigma^2 = 16$$

$$|X - 50| \geq 20$$

$$P(|X - 50| \geq 20) \leq \frac{16}{400} = \frac{1}{25}$$

Result

$$P(|X - 50| \geq 20) \leq 0.04$$

(*Slides: pages 48–52*)
L21_L22

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7. Chernoff Bound

Progression of Ideas

- Markov: bound using X

- Chebyshev: bound using $(X-\mu)^2$
- Chernoff: bound using e^{tX}
  (*Slide: page 54*)
  L21_L22

Key Insight

- Exponential grows very fast
- Large values of X become much larger
- Tail events get amplified
  (*Slides: pages 55–58*)
  L21_L22

Core Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

(*Slide: page 59*)
L21_L22

Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

- $e^{tX} \geq 0 \Rightarrow$ Markov applies
  (*Slide: page 61*)
  L21_L22

Final Form

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight

- Holds for all $t > 0$
- Choose best $t \Rightarrow$ tightest bound
- Much stronger than Markov/Chebyshev
- Exponential decay
  (*Slides: pages 62–64*)
  L21_L22

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8. Chernoff Bound (Lower Tail)

Goal

$$P(X \leq a)$$

Transformation

Use $t < 0$:

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

Final Form

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

(*Slides: pages 65–68*)
L21_L22

---

9. Chernoff Bound for Poisson Tail

Goal

Bound tail of:

$$X \sim \text{Poisson}(\lambda)$$

Step 1: Compute $E[e^{tX}]$

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \cdot e^{-\lambda} \frac{\lambda^i}{i!}$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

(*Slides: pages 69–70*)
L21_L22

End of Lecture 21–22

Sources